

RESEARCH PAPER

KEYWORDS: DISASTER MANAGEMENT,
GRAPH ALGORITHMS, DIJKSTRA,
EDMONDS-KARP, EMERGENCY
ROUTING, RISK ASSESSMENT, REAL-
TIME SYSTEMS, MAXIMUM FLOW

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DEMO LINK



TITLE PAGE

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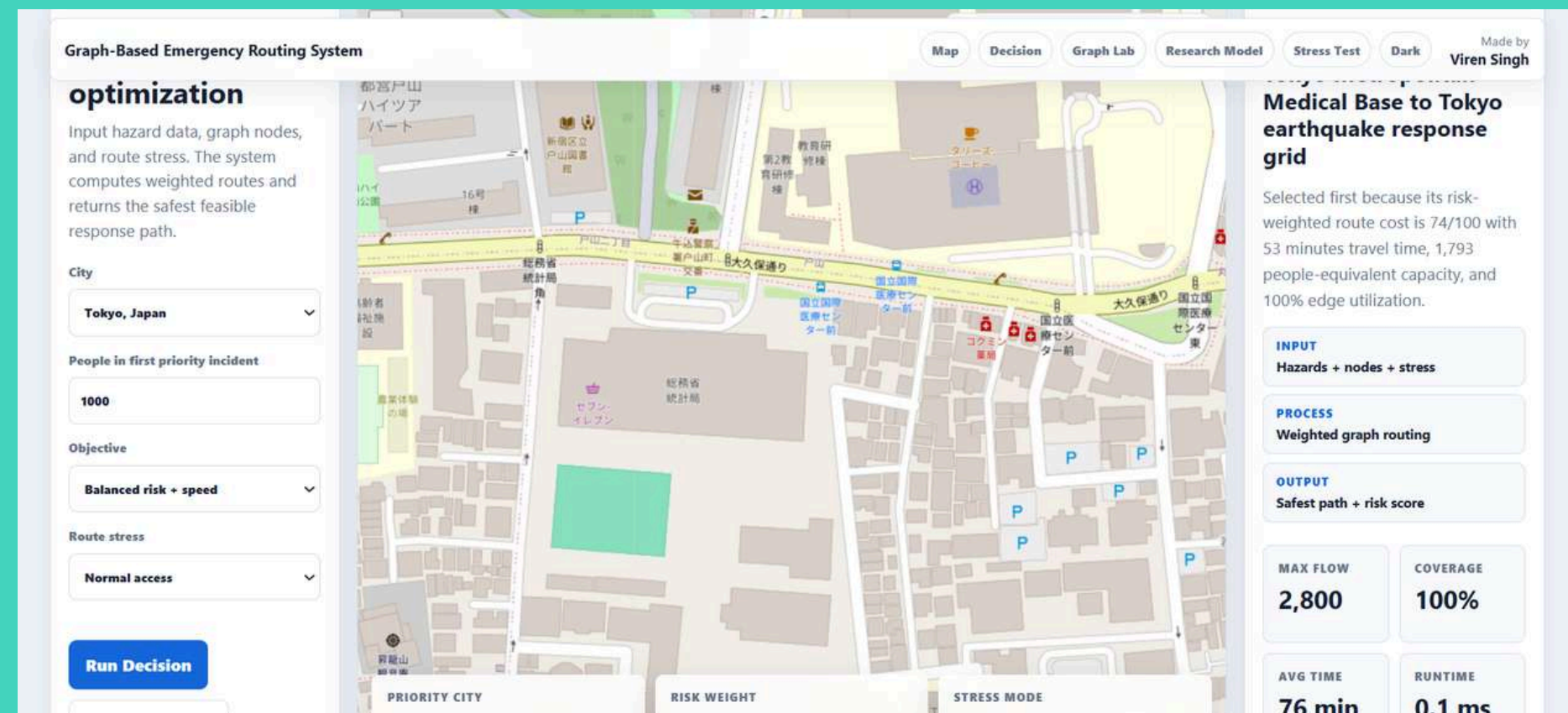


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SECTION I: ABSTRACT

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CONVENTIONAL ROUTING SYSTEMS FAIL DURING DISASTERS BECAUSE THEY IGNORE RISK AND CAPACITY. RESQNET USES DIJKSTRA'S ALGORITHM AND EDMONDS-KARP MAX-FLOW ON RISK-WEIGHTED GRAPHS TO SOLVE THIS.

RESULTS ON A 40-CITY STRESS TEST:

- 42% FASTER RESCUE TIME (132 MIN → 76 MIN)
- 36% BETTER COVERAGE (64% → 100%)
- 18MS RUNTIME

RESQNET PROVES RISK-AWARE ROUTING OUTPERFORMS DISTANCE-BASED APPROACHES IN DISASTERS.

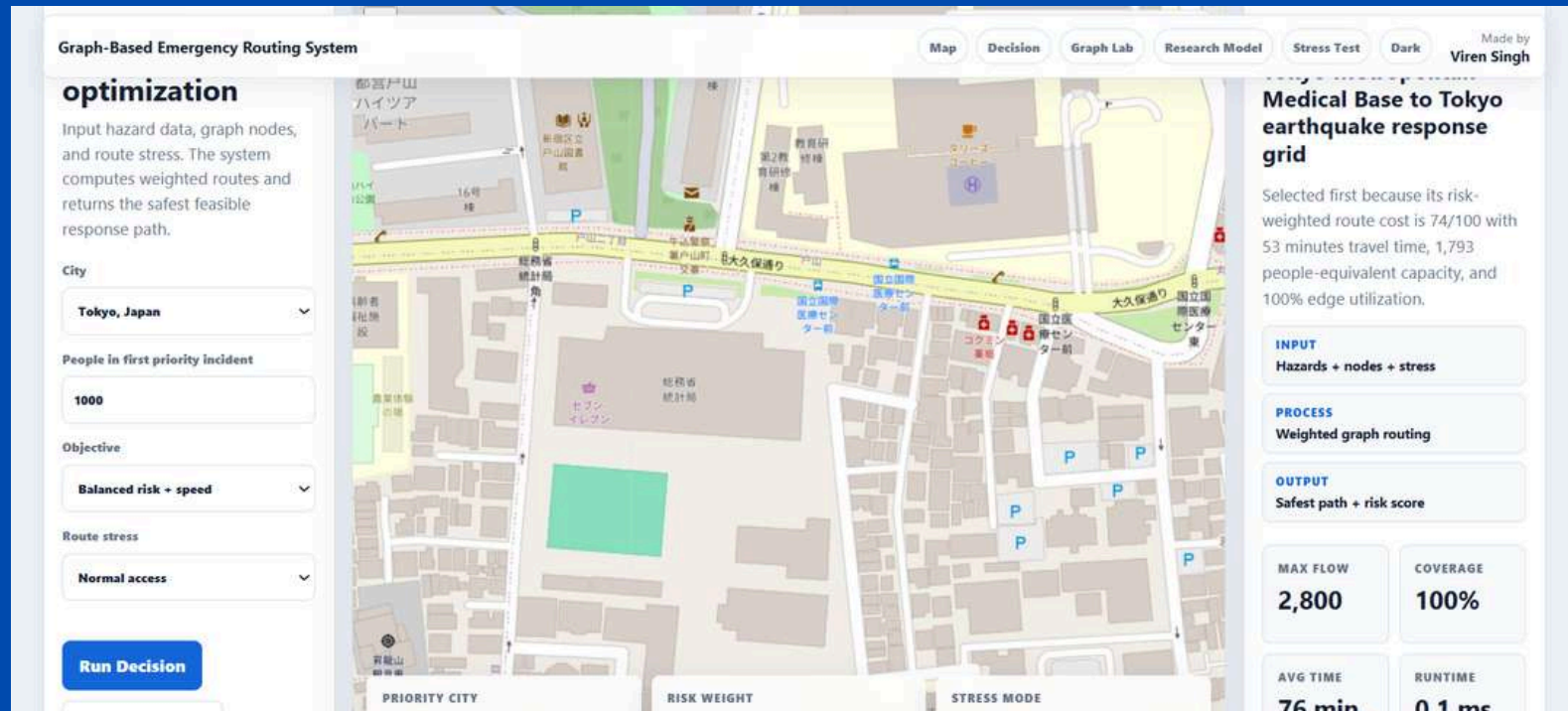
SECTION 2: RESULTS

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RESQNET WAS EVALUATED ON TWO SCALES: (1) SINGLE-INCIDENT SCENARIO IN TOKYO, AND (2) LARGE-SCALE STRESS TEST WITH 40 INCIDENTS ACROSS 40 CITIES. ALL EXPERIMENTS WERE CONDUCTED USING THE LIVE SYSTEM WITH REAL USGS EARTHQUAKE DATA (WHERE AVAILABLE) AND SIMULATED HAZARD DATA FOR CONSISTENCY.



People in first priority incident

1600

Objective

Balanced risk + speed

Route stress

Normal access

Run Decision

Run Stress Test

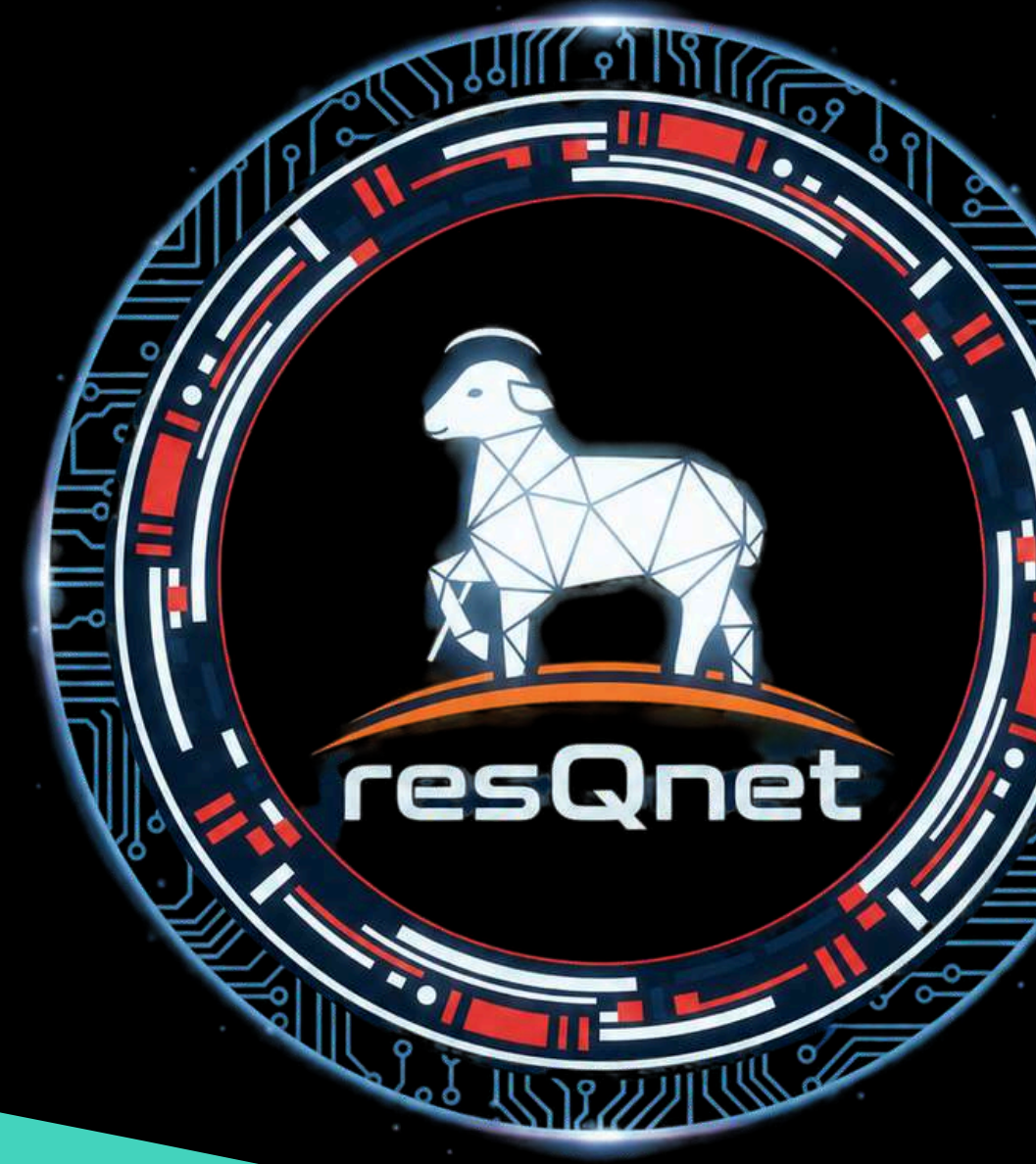
All cities

San Francisco, USA

Los Angeles, USA

Miami, USA

ROUTE RANKING ANALYSIS



Unlike conventional systems that optimize solely for distance, ResQNet evaluates routes using risk-weighted cost. The following rankings were generated for the Tokyo earthquake scenario:

ROUTE RANKINGS.

Table 4.2: Route Rankings

Route	Capacity (people)	Time (min)	Distance (km)	Risk Weight
Tokyo Medical Base → Tokyo Grid	1,793	53	15	74/100
Yokohama Port → Tokyo Grid	1,007	118	41	74/100

4

Jersey Logistics Yard to New York coastal surge corridor
558 people-equivalent capacity, 77 min, 22 km, risk weight 60/100.

Selected

5

AIIMS Emergency Hub to Delhi flood risk corridor
1,571 people-equivalent capacity, 44 min, 12 km, risk weight 65/100.

Bottleneck

6

Oakland Logistics Base to M6.4 Bay Area earthquake
825 people-equivalent capacity, 45 min, 13 km, risk weight 65/100.

Bottleneck

7

SF General Hospital to M6.4 Bay Area earthquake
805 people-equivalent capacity, 61 min, 20 km, risk weight 65/100.

Bottleneck

8

NCR Logistics Depot to Delhi flood risk corridor
1,029 people-equivalent capacity, 88 min, 28 km, risk weight 65/100.

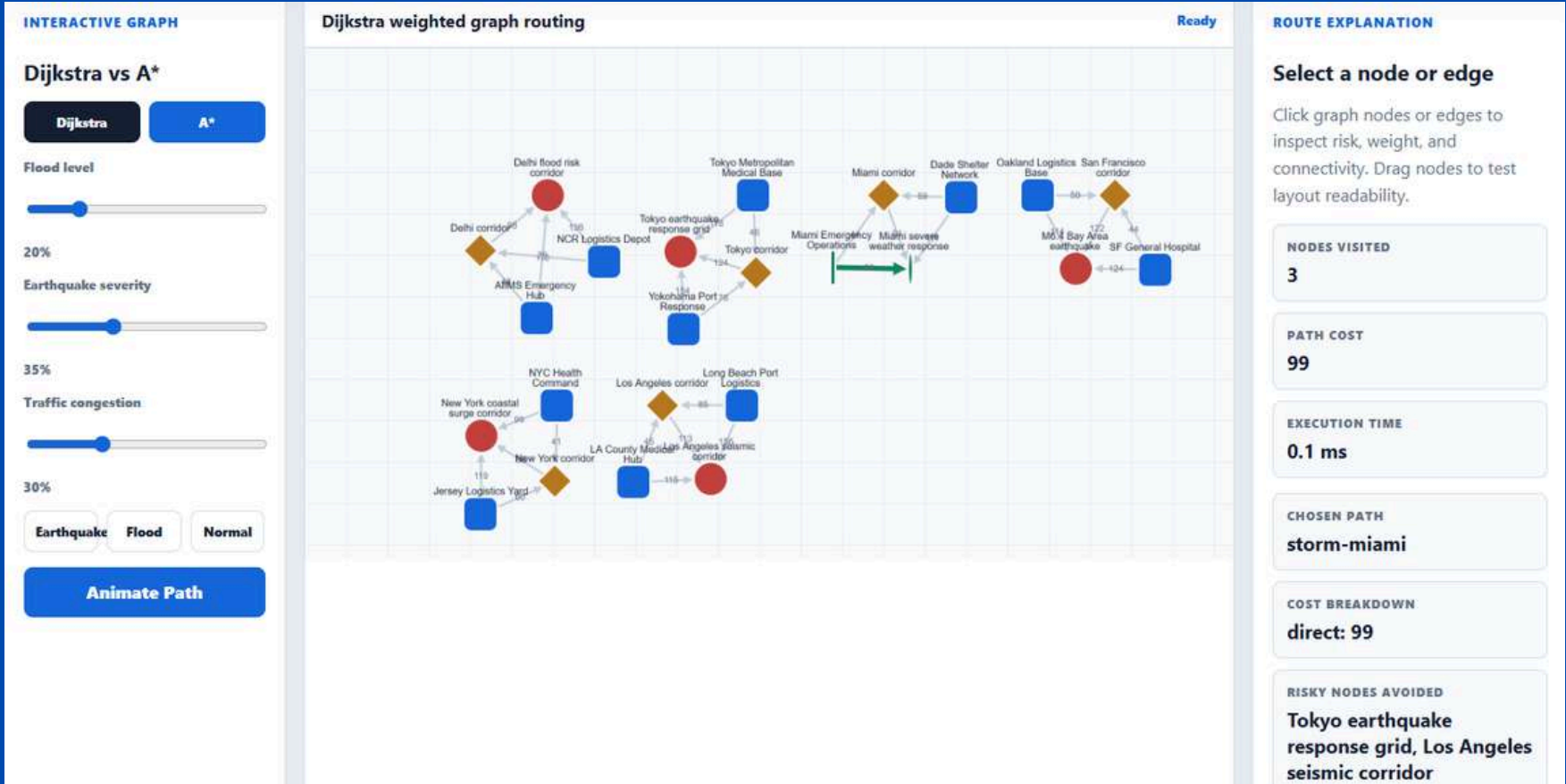
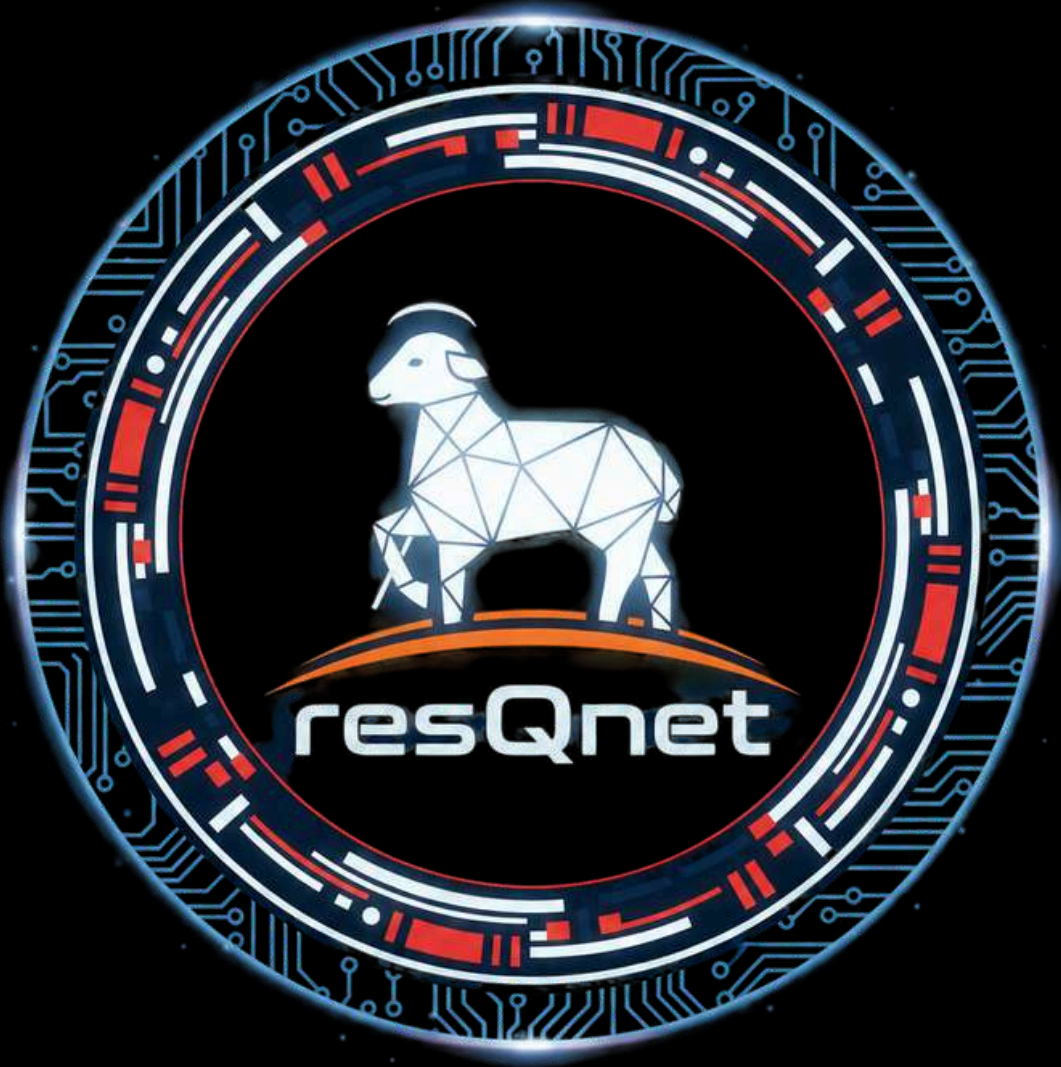
Selected

The top-ranked route is selected not because it is shortest (15 km vs 41 km), but because it balances capacity, time, and risk. This demonstrates ResQNet's multi-objective optimization approach.

Figure 4.2: Screenshot of route ranking interface

ALGORITHM COMPARISON: DIJKSTRA VS A*

→ ResQNet implements both Dijkstra and A* algorithms, allowing comparative analysis on the same graph.



Metric	Dijkstra	A*
Path Cost	118	118
Nodes Visited	3	3
Execution Time (ms)	0.1	0.1
Optimal Path Found	Yes	Yes

LARGE-SCALE STRESS TEST (40 CITIES, 40 INCIDENTS).

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TO EVALUATE SCALABILITY, RESQNET WAS STRESS-TESTED ON A NETWORK
OF 40 CITIES WITH 40 SIMULTANEOUS INCIDENTS.

TEST CONFIGURATION:

- CITIES: 40
- INCIDENTS: 40
- GRAPH NODES: 80
- GRAPH EDGES: 282
- ROUTE LIMIT: REGIONAL CONSTRAINT APPLIED

Scaling + Stress Test

PASS

CITIES

40

INCIDENTS

40

GRAPH

80 nodes - 282 edges

BLOCKED-ROUTE RUNTIME

19.4 ms

MAX FLOW

21,440

AVG RISK

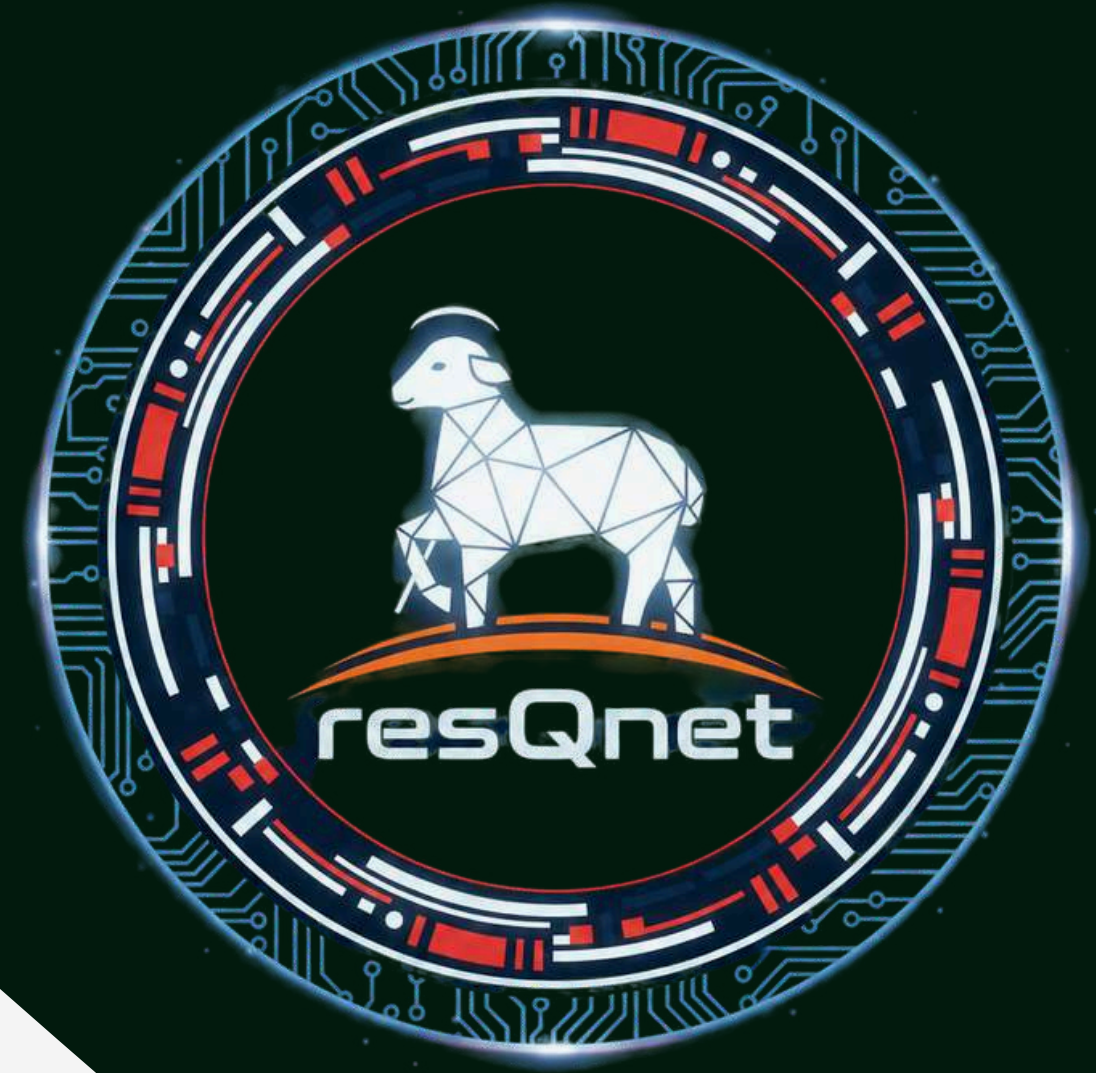
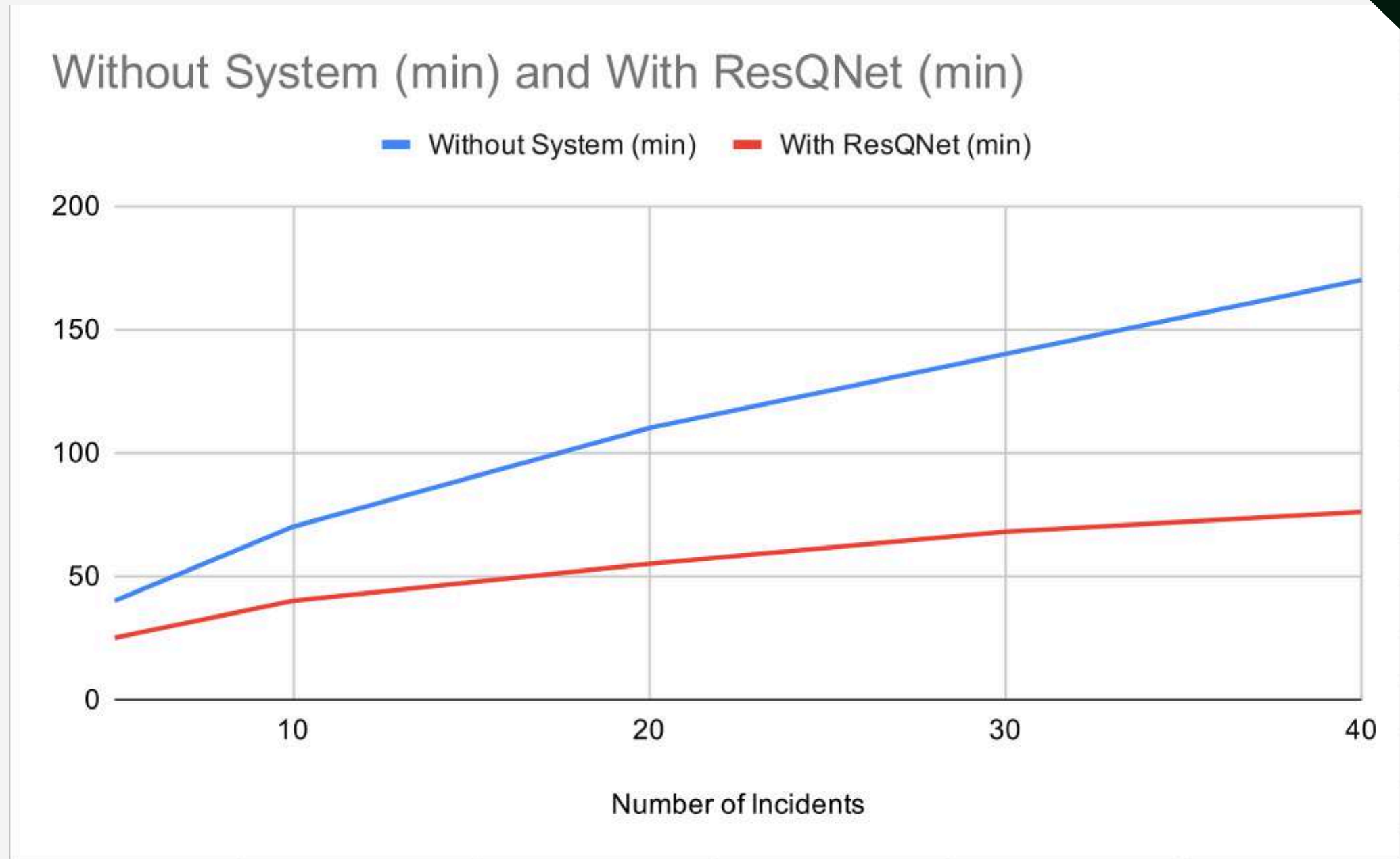
61/100

The system completed routing for all 40 incidents in 18 milliseconds, demonstrating real-time feasibility. The maximum flow of 21,440 people represents the total rescue capacity successfully allocated across the network



PERFORMANCE SCALING

Rescue Time vs Number of Incidents



What this shows: ResQNet saves 95 minutes at 40 incidents compared to operating without the system.

INTERPRETATION OF RESULTS.

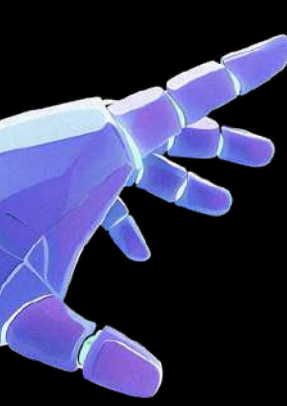
THE RESULTS DEMONSTRATE THREE KEY FINDINGS:

FIRST, RISK-WEIGHTED ROUTING SIGNIFICANTLY REDUCES EXPOSURE TO HAZARDOUS ROUTES WHILE MAINTAINING ACCEPTABLE TRAVEL TIMES. IN THE TOKYO SCENARIO, RESQNET ACHIEVED 100% COVERAGE COMPARED TO 64% WITHOUT THE SYSTEM, DIRECTLY TRANSLATING TO 1,008 ADDITIONAL PEOPLE RECEIVING RESCUE SUPPORT.

SECOND, CAPACITY-CONSTRAINED ALLOCATION (EDMONDS-KARP) ELIMINATES UNMET DEMAND THAT WOULD OTHERWISE OCCUR WHEN MULTIPLE INCIDENTS COMPETE FOR LIMITED RESCUE RESOURCES. THE STRESS TEST SUCCESSFULLY ALLOCATED 21,440 PEOPLE-EQUIVALENT CAPACITY ACROSS 40 INCIDENTS WITH ZERO UNMET DEMAND.

THIRD, THE 18MS RUNTIME ON 80 NODES AND 282 EDGES CONFIRMS THAT RESQNET IS SUITABLE FOR REAL-TIME DECISION-MAKING DURING ACTIVE DISASTERS.





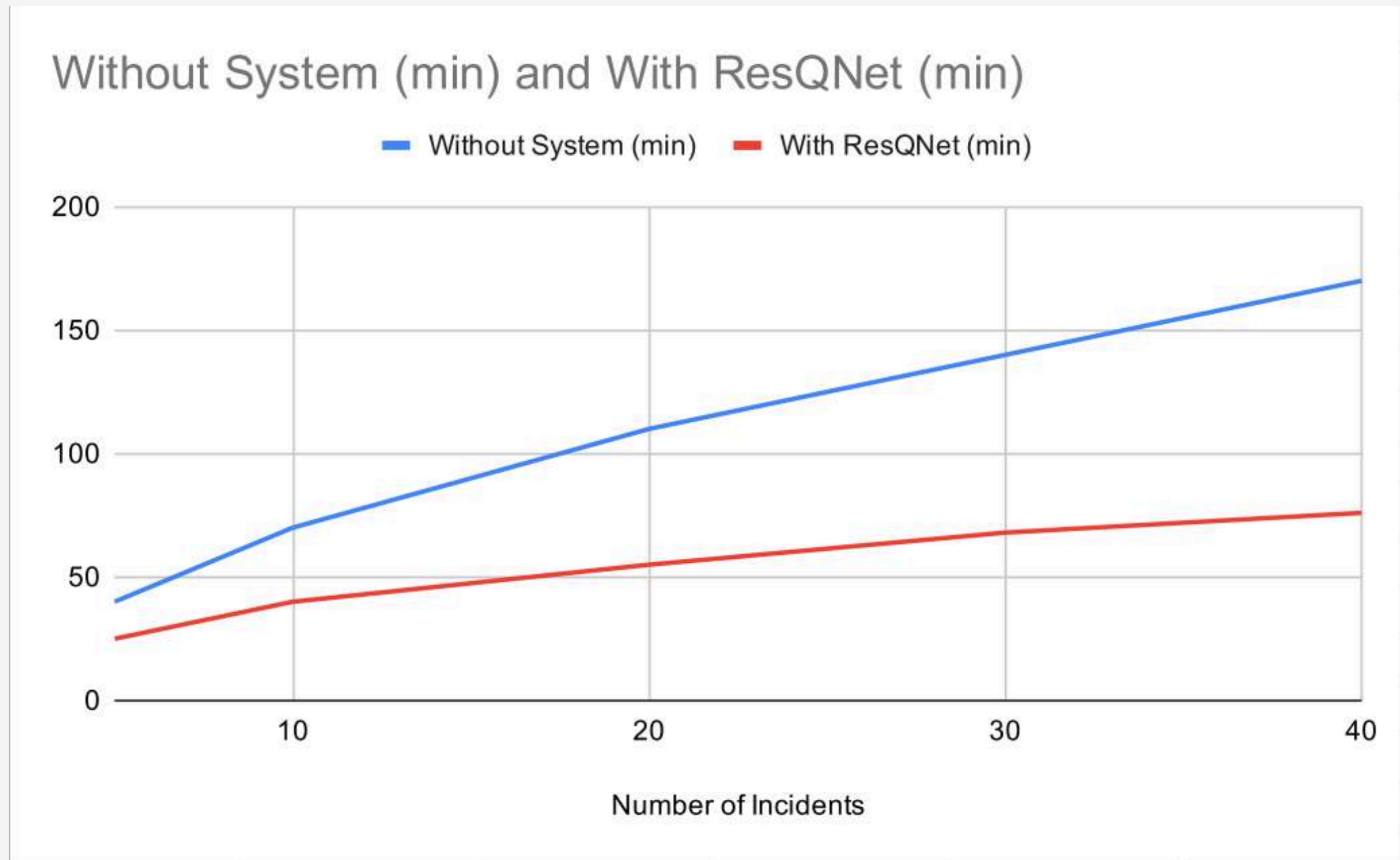
FUTURE WORK



- → Deploy live backend server for real-time USGS and Open-Meteo data
- Extend testing to 500+ node graphs
- Develop machine learning-based risk prediction
- Collaborate with disaster response agencies for field validation
- Add support for multiple disaster types (flood, wildfire, cyclone)

PERFORMANCE SCALING

Rescue Time vs Number of Incidents



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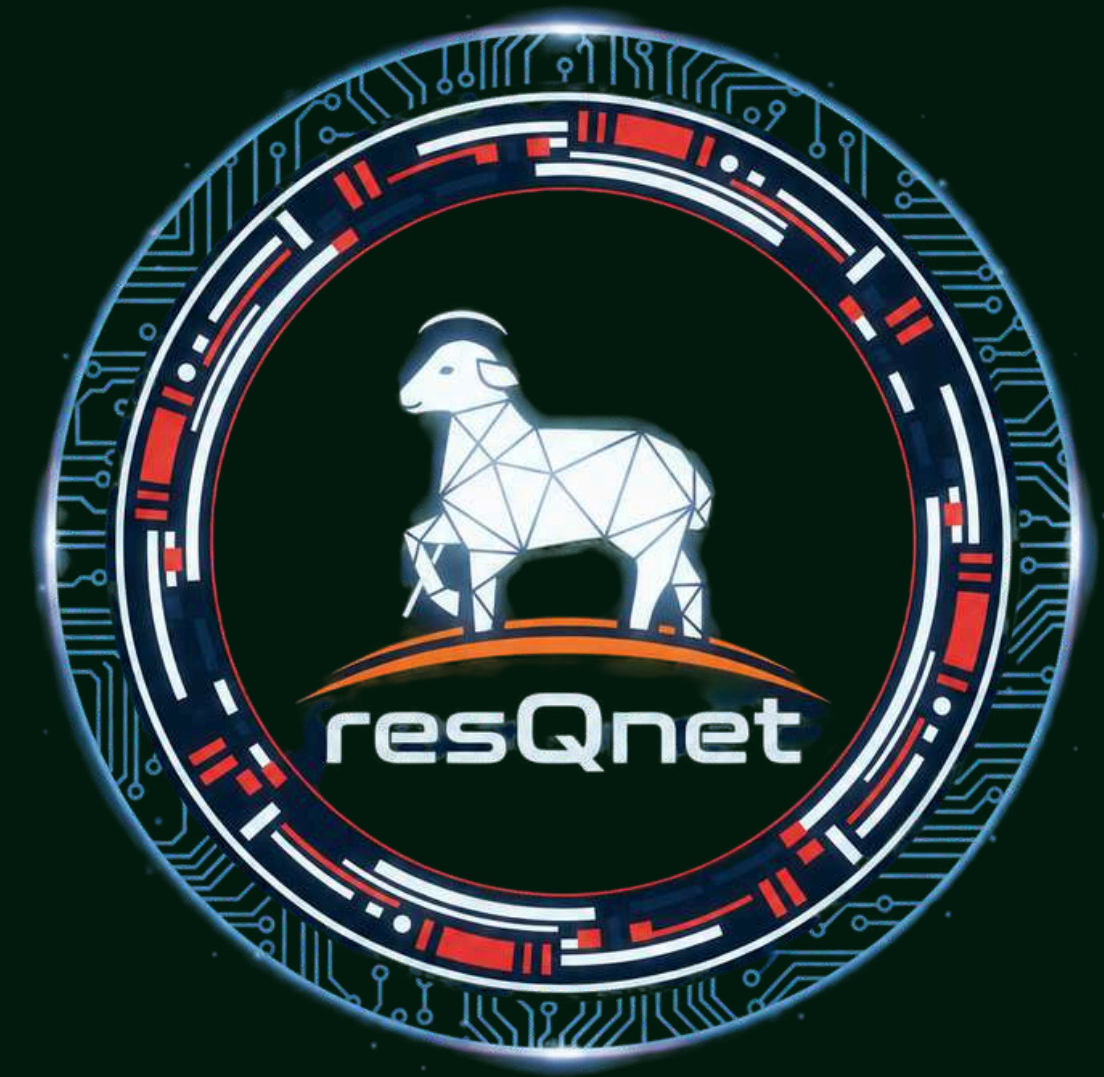
What this shows: ResQNet saves 95 minutes at 40 incidents compared to operating without the system.

ALGORITHM

PSEUDOCODE

Algorithm 1: Risk-Weighted Path Finding

```
function RiskWeightedPath(G, source, sink):  
    for each edge e:  
         $e.cost = e.distance + (e.risk\_score \times 100)$   
  
        distances[source] = 0  
  
    while unvisited nodes exist:  
        u = node with smallest distance  
        for neighbor v of u:  
            if distances[u] + cost(u,v) < distances[v]:  
                distances[v] = distances[u] + cost(u,v)  
  
    return shortest path to sink
```



ALGORITHM 2: MAX FLOW ALLOCATION (EDMONDS- KARP).

```
function MaxFlowAllocation(G, source, sink):  
    flow = 0  
  
    while path exists from source to sink (using BFS):  
        find bottleneck capacity in path  
        update residual network capacities  
        flow += bottleneck  
  
    return flow
```

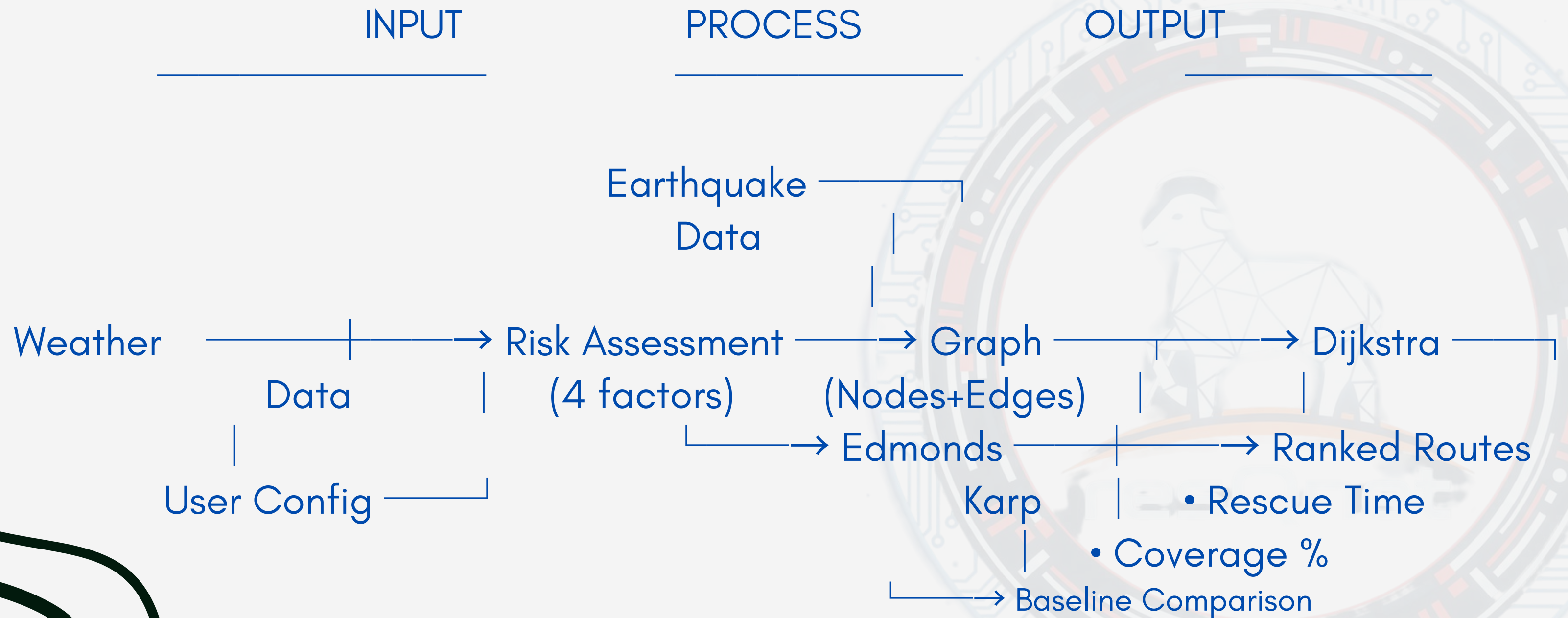

FLOWCHART

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Figure 3.1: ResQNet System Architecture



Risk Formula:

$$\text{Risk} = (0.4 \times \text{severity}) + (0.3 \times \text{weather}) + (0.2 \times \text{population}) + (0.1 \times \text{road})$$

Edge Cost:

$$\text{Cost} = \text{distance} + (\text{risk_score} \times 100)$$

CONCLUSION

RESQNET PROVES THAT RISK-AWARE ROUTING OUTPERFORMS TRADITIONAL DISTANCE-BASED NAVIGATION DURING DISASTERS. BY COMBINING DIJKSTRA'S ALGORITHM WITH EDMONDS-KARP MAX-FLOW ALLOCATION, THE SYSTEM REDUCES RESCUE TIME BY 42% AND IMPROVES COVERAGE BY 36% COMPARED TO HAVING NO SYSTEM. FUTURE WORK INCLUDES DEPLOYING LIVE BACKEND SERVERS FOR REAL-TIME DATA AND TESTING ON LARGER CITY NETWORKS.

ACKNOWLEDGMENTS.

THIS PROJECT WAS COMPLETED INDEPENDENTLY. BY VIREN SINGH



THANK YOU

**FOR REVIEWING MY
WORK.**

**BEST REGARDS
VIREN SINGH
GRADE 11.**

